Patient Record — ID 1823

Age	28	**Sex**	F
Race/Ethnicity	Cypriot (Mediterranean)	**Insurance**	Commercial PPO
Total Encounters	8	**Generation Seed**	5615

Primary Diagnoses

• Thalassemia minor
• Endometriosis of ovary (Ovarian endometrioma)
• Iron deficiency anemia

Comorbidities

• Menorrhagia
• Family history of hereditary bleeding disorder

Patient Profile

Sex	F
Age at First Encounter	28
Race/Ethnicity	Cypriot (Mediterranean)
Insurance	Commercial PPO
Occupation	Elementary school teacher
Smoking Status	Never
Alcohol Use	Social (1-2 drinks per month)

Allergies:

• NKDA

Chronic Conditions:

• Beta-thalassemia minor (trait)
• Endometriosis
• History of iron deficiency anemia

Home Medications:

• Multivitamin - 1 tablet daily

Surgical History:

• Laparoscopic right ovarian cystectomy for endometrioma

Family History:

• Father with beta-thalassemia minor
• Mother with beta-thalassemia minor
• Paternal cousin with beta-thalassemia major
• Maternal aunt with endometriosis

Encounters (8)

Encounter 1 — 2020-01-15 — Internal Medicine

Encounter ID	7659	**Type**	outpatient
Attending	Dr. Sarah Chen	**Order**	0
Department	Internal Medicine	**Gen. Method**	template

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Evaluation of anemia noted during a pre-employment physical for elementary school teaching position.

HPI

She feels well, denying fatigue, dyspnea, or dizziness. She reports regular menstrual cycles and no history of heavy bleeding. Her diet includes red meat, leafy greens, and iron-fortified cereals. She is not currently pregnant but plans to start a family within the next year. She is started on oral ferrous sulfate but returns six weeks later with no improvement.

PMH (modified)

Active Problem List:
- Beta-thalassemia minor (trait)
- Endometriosis
- History of iron deficiency anemia

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS

She takes a daily multivitamin and does not take any medications or supplements.

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

Her mother and maternal uncle were both told they had "small blood cells" but never required treatment.

SOCIAL_HISTORY

She emigrated from Cyprus six years ago and works as a schoolteacher.

PHYSICAL_EXAM

Physical exam is normal.

LABS

Laboratory studies show microcytic anemia. Iron panel reveals normal iron and ferritin levels. Hemoglobin electrophoresis is ordered, and results show elevated hemoglobin A2 with mild increase in hemoglobin F.

Encounter 2 — 2020-04-15 — Hematology & Oncology

Encounter ID	7656	**Type**	outpatient
Attending	Dr. Michael Rodriguez	**Order**	1
Department	Hematology & Oncology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Increasing fatigue over the past two months with occasional lightheadedness on standing.

HPI (modified)

The patient returns for evaluation of progressive fatigue and lightheadedness, following her initial assessment for anemia detected during a pre-employment physical three months prior, with known Beta-thalassemia minor and Endometriosis. She reports feeling tired during her daily activities and occasionally feeling lightheaded when standing up quickly. Her menstrual cycles are regular but tend to be heavy, requiring frequent pad changes. She denies any unusual cravings or gastrointestinal symptoms.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Beta-thalassemia minor (trait)
- Endometriosis
- History of iron deficiency anemia

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

She does not take any medications

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

There is no family history of blood disorders.

SOCIAL_HISTORY

Follows a balanced diet

PHYSICAL_EXAM

Physical examination shows pale conjunctivae but is otherwise normal.

LABS

Laboratory results:
Hemoglobin 9.4 g/dL
MCV 72 fL
Serum iron 24 µg/dL
Ferritin 9 ng/mL

RDW 17%

OTHER_STUDIES

A peripheral blood smear is obtained and shown below.

Encounter 3 — 2020-09-15 — Obstetrics & Gynecology

Encounter ID	7655	**Type**	procedure
Attending	Dr. Emily Johnson	**Order**	2
Department	Obstetrics & Gynecology	**Gen. Method**	template

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Pelvic pain and evaluation for ovarian cyst.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Beta-thalassemia minor (trait)
- Endometriosis

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father with beta-thalassemia minor
- Mother with beta-thalassemia minor
- Paternal cousin with beta-thalassemia major
- Maternal aunt with endometriosis

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

Encounter 4 — 2021-03-15 — Internal Medicine

Encounter ID	7654	**Type**	outpatient
Attending	Dr. Sarah Chen	**Order**	3
Department	Internal Medicine	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Routine health examination and follow-up of chronic anemia.

HPI (modified)

Returning for anemia follow-up with thalassemia minor, iron deficiency, and endometriosis, she has a history of mild fatigue and denies any recent illness, bleeding, or dietary changes.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Endometriosis of ovary (Ovarian endometrioma) (diagnosed 2020-09-15)
- Beta-thalassemia minor (trait)
- Endometriosis

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

Her family history is notable for a sister with a "blood disorder" that was never clearly explained to her.

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

PHYSICAL_EXAM

On physical examination, she appears well and has no pallor, jaundice, or organomegaly.

LABS

Her complete blood count reveals the following:

- Hemoglobin: 11.2 g/dL (Normal: 12-16 g/dL)
- Mean corpuscular volume (MCV): 72 fL (Normal: 80-100 fL)
- Red cell distribution width (RDW): 13.5% (Normal: 11.5-14.5%)
- Reticulocyte count: 3.5% (Normal: 0.5-2.5%)

Encounter 5 — 2022-03-15 — Hematology & Oncology

Encounter ID	7652	**Type**	outpatient
Attending	Dr. Michael Rodriguez	**Order**	4
Department	Hematology & Oncology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Three-month history of worsening fatigue and shortness of breath on exertion.

HPI (modified)

Returning for follow-up of beta-thalassemia minor, iron deficiency anemia, and endometriosis, she reports heavy menses but otherwise feels well.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Endometriosis of ovary (Ovarian endometrioma) (diagnosed 2020-09-15)
- Beta-thalassemia minor (trait)
- Endometriosis

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

She was started on a daily iron supplement at the time.

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father with beta-thalassemia minor
- Mother with beta-thalassemia minor
- Paternal cousin with beta-thalassemia major
- Maternal aunt with endometriosis

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

LABS

Hemoglobin is 9.2 g/dL, MCV is 72 fL. One month ago, her hemoglobin was 9.0 g/dL and MCV was 73 fL.

Encounter 6 — 2023-03-15 — Hematology & Oncology

Encounter ID	7653	**Type**	outpatient
Attending	Dr. David Park	**Order**	5
Department	Hematology & Oncology	**Gen. Method**	template

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Ongoing complaints of fatigue and mild exertional dyspnea.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Endometriosis of ovary (Ovarian endometrioma) (diagnosed 2020-09-15)
- Beta-thalassemia minor (trait)
- Endometriosis

She denies any significant past medical history.

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

She reports a similar history of anemia in several family members, but she is not aware of any specific diagnosis

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

PHYSICAL_EXAM

On physical examination, she appears well with no pallor, jaundice, or organomegaly

LABS

Her complete blood count reveals:

- Hemoglobin: 11.2 g/dL
- Mean corpuscular volume (MCV): 68 µm^3
- Red blood cell (RBC) count: 6.1 x 10^12/L
- Platelets: 250 x 10^9/L
- White blood cells (WBC): 6.0 x 10^9/L

OTHER_STUDIES

A peripheral blood smear shows target cells.

PLAN

Serum iron studies and hemoglobin electrophoresis are ordered.

Encounter 7 — 2024-03-15 — Hematology & Oncology

Encounter ID	7658	**Type**	outpatient
Attending	Dr. David Park	**Order**	6
Department	Hematology & Oncology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Evaluation of low hemoglobin discovered during recent blood donation attempt.

HPI (modified)

The patient presents for follow-up evaluation of anemia in the context of longstanding beta-thalassemia minor, with prior hematology encounters for similar complaints. She reports symptoms have been present for years and were previously attributed to "low iron," for which she briefly took supplements in the past. She recently donated blood and was informed her hemoglobin was low. She denies gastrointestinal symptoms, heavy menstrual bleeding, or recent infections.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Endometriosis of ovary (Ovarian endometrioma) (diagnosed 2020-09-15)
- Beta-thalassemia minor (trait)
- Endometriosis

She has no chronic diseases.

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

Her family history is significant for a cousin with "some kind of blood disorder."

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

PHYSICAL_EXAM

Physical examination is unremarkable, with no pallor, jaundice, or splenomegaly.

LABS

Laboratory studies show:

Hemoglobin	10.5 g/dL
Hematocrit	31%
Erythrocyte count	5.9 million/mm³

MCV	70 µm³
RDW	13.1% (normal: 13–15%)
Reticulocyte count	1.1%
Serum ferritin	118 ng/mL
Serum iron	85 µg/dL
TIBC	260 µg/dL
Coombs test	Negative

Encounter 8 — 2025-03-15 — Hematology & Oncology

Encounter ID	7657	**Type**	outpatient
Attending	Dr. Priya Patel	**Order**	7
Department	Hematology & Oncology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Asymptomatic participation in hemoglobin-oxygen dissociation research study.

HPI (modified)

The patient with known beta-thalassemia minor returns for annual hematology follow-up. She is evaluated as part of a study for hemoglobin-oxygen dissociation metrics.

PMH (modified)

Active Problem List:
- Thalassemia minor (diagnosed 2020-01-15)
- Iron deficiency anemia (diagnosed 2020-04-15)
- Endometriosis of ovary (Ovarian endometrioma) (diagnosed 2020-09-15)
- Beta-thalassemia minor (trait)
- Endometriosis

PSH (modified)

- Laparoscopic right ovarian cystectomy for endometrioma

MEDICATIONS (modified)

Home Medications:
- Multivitamin 1 tablet daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father with beta-thalassemia minor
- Mother with beta-thalassemia minor
- Paternal cousin with beta-thalassemia major
- Maternal aunt with endometriosis

SOCIAL_HISTORY (modified)

Smoking: Never. Alcohol: Social (1-2 drinks per month). Occupation: Elementary school teacher.

LABS

Curve is slightly left-shifted, with an increase in HbF and HbA2